Timeline Check-In

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Set Up

Packages

```
# load in tidyverse package
library(tidyverse)
# load in here package
library(here)
# load in janitor package
library(janitor)
# load in snakecase package
library(snakecase)
# load in scales package
library(scales)
# load in readxl package
library(readxl)
# load in naniar package
library(naniar)
# load in patchwork package
library(patchwork)
# load in dplyr package
library(dplyr)
# load in broom package
library(broom)
# load in corrplot package
library(corrplot)
# load in ggrepel package
library(ggrepel)
```

Data

```r
# read in taxonomic info data
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# read in aquatic ivert. data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# read in site data
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# read in water quality data
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Creating Consistent Aesthetics

```r
# shared color palette for taxon-based figures
taxon_colors <- c(
  "Copepod" = "#1B9E77",
  "Ostracod" = "#D95F02",
  "Cladocera" = "#7570B3",
  "Hemiptera Corixidae Boatman" = "#E7298A",
  "Nematode" = "#66A61E"
)

# shared color palette for DO-category figures
do_colors <- c(
  "low" = 'red3',
  "moderate" = "white",
  "high" = 'blue3')

# shared colors for model lines, summary points, and raw points
model_color <- "#4C78A8"
summary_color <- "#222222"
raw_point_color <- "#333333"
ci_fill <- "#A6CEE3"
reference_color <- "gray45"
```

```r
# shared theme for all figures
theme_ncos <- theme_classic() +
  theme(
    plot.title = element_text(face = "bold", size = 13),
    axis.title = element_text(size = 11),
    axis.text = element_text(size = 9),
    strip.background = element_rect(fill = "gray90", color = "gray50"),
    strip.text = element_text(face = "bold"),
    legend.position = "right"
  )
```

Cleaning

`sites` object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning `taxon_list`

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

Cleaning `water_quality`

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
```

```r
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning `aq_ins`

```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
```

```r
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

Focus Taxa (`focus_taxa`)

```r
# taxa to include in all figures
focus_taxa <- c("Ostracod", "Copepod", "Cladocera",
                "Hemiptera Corixidae Boatman", "Nematode")
```

Creating `DO_taxa`, `total_abundance`, and DO bins

```r
# create a new objecy aq_ins_clean_bins that adds DO bins to aq_ins_clean
aq_ins_clean_bins <- aq_ins_clean |>
  # create a new bins column based on site_full values
```

```r
  mutate(bins = case_when(site_full == "Phelps Road" ~ "low",
                           site_full == "North of Phelps Bridge" ~ "low",
                           site_full == "South of Venoco Bridge" ~ "moderate",
                           site_full == "South of Phelps Bridge" ~ "moderate",
                           site_full =="Devereux Creek" ~ "moderate",
                           site_full =="North of Venoco Bridge" ~ "high",
                           site_full == "Main Channel" ~ "high")) |>
  # replace underscores in taxon with spaces & convertto title case
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # keep only taxa included in the focus_taxa object
  filter(taxon %in% focus_taxa) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert bins into an ordered factor
  mutate(bins = factor(bins, levels = c("low", "moderate", "high")))

# create object DO_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
  # remove rows that are missing median salinity values
  filter(!is.na(med_do)) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # mutates to title-case taxon names
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # only include focus taxa
  filter(taxon %in% focus_taxa) |>
  # order taxon levels by maximum average abundance per liter
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep only the columns needed for the figure
  select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)

# create objetc total_abundance from DO_taxa
total_abundance <- DO_taxa |>
  # keep only taxa included in the focus_taxa object
  filter(taxon %in% focus_taxa) |>
  # group observations by date, site, and median DO
  group_by(date, site, med_do) |>
  # sum average abundance per liter across taxa & ignore missing values
  summarize(total_abundance = sum(ave_lit, na.rm = TRUE),
            # ungroup
            .groups = "drop") |>
  # create a log10-transformed total abundance column called log_total_abundance
```

```
  mutate(log_total_abundance = log10(total_abundance + 1))

# updates the total_abundance object
total_abundance <- total_abundance |>
  # create a new variable do_category based on DO quantiles
  mutate(
    # classify each observation into low, medium, or high dissolved oxygen
    do_category = case_when(
      # assign Low DO to values at or below the 25th percentile of median DO
      med_do <= quantile(med_do, 0.25, na.rm = TRUE) ~ "Low DO",
      # assign Medium DO to values between the 25th percentile - 75th percentile
      med_do <= quantile(med_do, 0.75, na.rm = TRUE) ~ "Medium DO",
      # assign High DO to all remaining values above the 75th percentile
      TRUE ~ "High DO"))

# edit the column do_category into an ordered factor
total_abundance$do_category <- factor(total_abundance$do_category,
                                      # set the category order
                                      levels = c("Low DO", "Medium DO", "High DO"))
```

Ideas for analysis and background

Ideas for analysis:

Our first big idea is looking at total invertebrate abundance compared to median dissolved oxygen levels. We ran a correlation test and found that there was a statistically significant negative linear relationship between median dissolved oxygen and log-transformed total invertebrate abundance, suggesting that abundance decreased as DO increased.

We also explored how dissolved oxygen levels differ between sites, and then categorized sites into low, moderate, and high DO bins. We found that total log-transformed invertebrate abundance did not differ significantly across low, moderate, and high DO bins. However, Ostracod abundance differed significantly across DO bins, indicating that Ostracods may respond differently to low, moderate, and high DO conditions. After Bonferroni correction, the moderate vs. high DO comparison was close to significant, suggesting a possible difference in Ostracod abundance between these bins but not strong enough to conclude significance at $= 0.05$.

Another way we compared taxa in bins was by running a Chi-square test on taxon composition within each DO bin. We found that taxon composition differed significantly across DO bins, suggesting that the invertebrate community structure changes depending on dissolved oxygen category. Next, we'd like to perform post-hoc tests to see which groups differ from one another.

Background:

North Campus Open Space is a restored wetland, which makes it a great area to explore data about the effects of restoration. One metric of restoration progress is water quality, and dissolved oxygen concentration is a key water-quality indicator. We are interested in the relationship between invertebrate abundance and dissolved oxygen, which ultimately ties back to the health of this wetland. This is a less traditional method of measuring ecosystem health, but can become more widely used with increasing data showing these relationships. This study applies to all wetland ecosystems and aquatic environments more broadly. In addition to being a metric of ecosystem health, invertebrate abundance compared to dissolved oxygen can tell us about the conditions that these invertebrates exist in.

Citations:

Poster: Algae, Macroinvertebrate and Water Quality Relationships at North Campus Open Space - link

Background Articles

Connolly, N. M., Crossland, M. R., & Pearson, R. G. (2004). *Effect of low dissolved oxygen on survival, emergence, and drift of tropical stream macroinvertebrates.* BioOne Complete (BioOne). https://doi.org/10.1899/0887-3593(2004)023%3C0251:eoldoo%3E2.0.co;2

Gunsch, D. E. (2020, November 5). *Community Composition of Aquatic Invertebrates along Dissolved Oxygen Gradients in Lake Huron Coastal Wetland Wet Meadows.* Handle.net. https://hdl.handle.net/20.500.14776/9073

Lucchesi, D. O., Chipps, S. R., & Schumann, D. A. (2022). *Effects of seasonal hypoxia on macroinvertebrate communities in a small reservoir.* Lakes & Reservoirs: Science, Policy and Management for Sustainable Use, 27(1). https://doi.org/10.1111/lre.12395

Suren, A. M., Lambert, P., & Sorrell, B. K. (2010). *The Impact of Hydrological Restoration on Benthic Aquatic Invertebrate Communities in a New Zealand Wetland.* Restoration Ecology, 19(6), 747–757. https://doi.org/10.1111/j.1526-100x.2010.00723.x

Spearman Correlation Analysis for Total Abundance and DO levels

```r
# run a Spearman correlation test between log total abundance & median DO
cor.test(
  # use log-transformed total abundance as the first variable
  total_abundance$log_total_abundance,
  # use median DO as the second variable
  total_abundance$med_do,
  # specify Spearman method
  method = "spearman",
  # use approximate p-values
  exact = FALSE)
```

```
	Spearman's rank correlation rho

data:  total_abundance$log_total_abundance and total_abundance$med_do
S = 8526.3, p-value = 0.08182
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.3027201
```

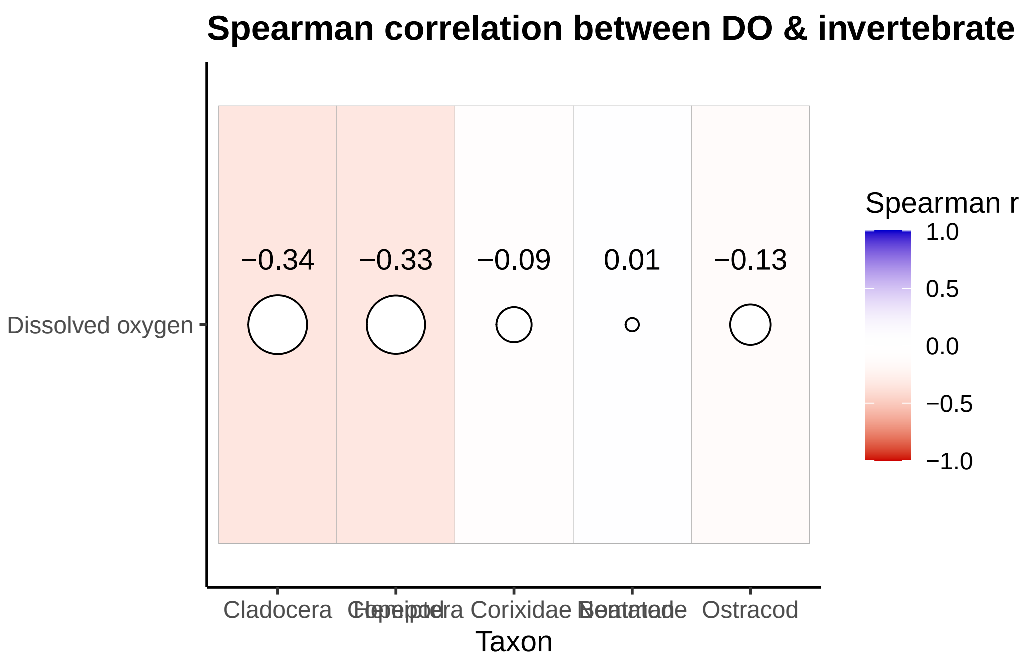
Total log-transformed invertebrate abundance had a **weak negative** relationship with dissolved oxygen, but this relationship was not statistically significant using Spearman correlation.

Correlation analysis for Each Focus Taxa and DO levels

```r
# create wide dataset (one column per taxon)
taxa_wide <- DO_taxa |>
  # keep only the columns needed
  select(date, site, taxon, log_ave_lit, med_do) |>
  # convert the dataset from long format to wide format
  pivot_wider(names_from = taxon, # create one new column for each taxon
              # fill each taxon column with log-transformed abundance values
              values_from = log_ave_lit)

# calculate correlation of each taxon with dissolved oxygen
# calculate Spearman correlation of each taxon with dissolved oxygen
cor_do <- taxa_wide |>
  # remove date and site columns so only DO and taxon abundance columns remain
```

```r
  select(-date, -site) |>
  # calculate correlations for each taxon column
  summarize(across(-med_do, ~ cor(.x,
                                  # correlate each column w/ median DO
                                  med_do,
                                  # specify spearman method
                                  method = "spearman",
                                  # only use rows w/ complete observations
                                  use = "complete.obs"))) |>
  # convert the correlation results from wide format to long format
  pivot_longer(cols = everything(),
               # store taxon names in a column called taxon
               names_to = "taxon",
               # store correlation values in a column called correlation
               values_to = "correlation")

ggplot(cor_do, # create a new plot from cor_do
       aes(x = taxon, # x-axis is taxon
           y = "Dissolved oxygen", # titles y-axis
           fill = correlation)) + # fill color by correlation
  # add tiles where fill represents the strength + direction of correlation
  geom_tile(color = "grey70") +
  # add circles sized by the absolute value of the correlation
  geom_point(aes(size = abs(correlation)), shape = 21, fill = "white") +
  # add labels above circles
  geom_text(aes(label = round(correlation, 2)),
            # move the text labels slightly above the circles
            nudge_y = 0.15, size = 4) +
  # create a color scale for negative, neutral, and positive correlations
  scale_fill_gradient2(
    # use red for negative correlations
    low = do_colors["low"],
    # use white for correlations near zero
    mid = do_colors["mid"],
    # use blue for positive correlations
    high = do_colors["high"],
    # sets midpoint to 0
    midpoint = 0,
    # set the color scale limits from -1 to 1
    limits = c(-1, 1),
    # labels legend 'Spearman r'
    name = "Spearman r") +
```

```r
  # set the range of circle sizes + remove the size legend
  scale_size(range = c(2, 10), guide = "none") +
  labs(x = "Taxon", # labels x-axis
       y = NULL, # remove the y-axis label
       # creates descriptive title
       title = "Spearman correlation between DO & invertebrate abundance") +
  # use consistent theme
  theme_ncos
```



Most focus taxa showed **weak negative** correlations with dissolved oxygen, with Cladocera and Copepod showing the strongest negative relationships and Nematode showing almost no relationship.

Pearson correlation for continuous DO and total community abundance

```r
# run a Pearson correlation test between log total abundance + median DO
cor.test(
  total_abundance$log_total_abundance, # log total abundance is first variable
  total_abundance$med_do, # use median DO as the second variable
         method = "pearson") # specify pearson method
```

```
	Pearson's product-moment correlation

data:  total_abundance$log_total_abundance and total_abundance$med_do
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.63289820 -0.04217141
sample estimates:
       cor 
-0.3749894 
```

```
# r = -0.3749894, p = 0.02887
# there is a significant negative correlation between abundance and continous DO
```

There was a **statistically significant negative linear relationship** between median dissolved oxygen and log-transformed total invertebrate abundance, suggesting that abundance decreased as DO increased.

Kruskal-Wallis test comparing abundance across DO categories

```
# run a Kruskal-Wallis test comparing log total abundance across DO categories
kruskal.test(log_total_abundance ~ do_category,
             # use the total_abundance dataset
             data = total_abundance)
```

```
	Kruskal-Wallis rank sum test

data:  log_total_abundance by do_category
Kruskal-Wallis chi-squared = 2.0001, df = 2, p-value = 0.3679
```

```
# Kruskal-Wallis chi-squared = 3.2212, df = 2, p-value = 0.1998
# there is not a significant difference between abundance and DO categories
```

Total log-transformed invertebrate abundance **did not differ significantly** across low, moderate, and high DO categories.

Chi-Square

```
# create abundance table for chi-square test
chi_square_table <- aq_ins_clean_bins |>
  # keep only rows with the focus taxa
  filter(taxon %in% focus_taxa) |>
  # group data by DO bin + taxon
  group_by(bins, taxon) |>
  summarise(
    # sum average abundance per liter + ignore missing values
    total_abundance = sum(ave_lit, na.rm = TRUE),
    # ungroup
    .groups = "drop") |>
  # convert the table from long format to wide format
  pivot_wider(
    # create one column for each taxon
    names_from = taxon,
    # fill taxon columns with total abundance values
    values_from = total_abundance,
    # fill missing values with 0
    values_fill = 0) |>
  # convert the bins column into row names
  column_to_rownames("bins") |>
  # convert the table into a matrix
  as.matrix()

# show table
chi_square_table
```

```
         Cladocera   Copepod Hemiptera Corixidae Boatman  Nematode    Ostracod
low       980.2667  115.8667                   0.2666667 0.1333333 1202.933333
moderate  510.6667 1170.4000                  83.8666667 0.2666667 1057.866667
high      264.2667 1453.4667                   8.8000000 0.0000000    7.866667
```

```
# chi-square test for differences in taxon composition across DO bins
chi_square_test <- chisq.test(chi_square_table)

# show chi-square result
chi_square_test
```

```
	Pearson's Chi-squared test

data:  chi_square_table
X-squared = 2859.2, df = 8, p-value < 2.2e-16
```

Taxon composition **differed significantly across DO bins**, suggesting that the invertebrate community structure changes depending on dissolved oxygen category.

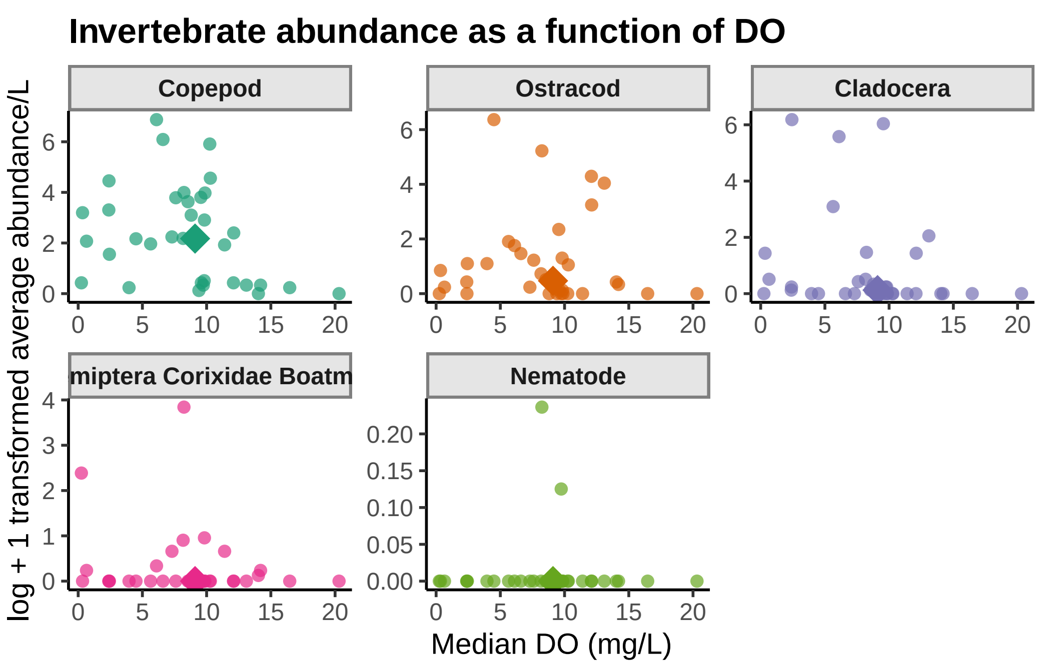
Visualizations directly relevant to answering questions

Directly Relevant Figure 1: DO levels vs invert. abundance for each focus taxa

```r
# creates object DO_taxa_medians from DO_taxa
DO_taxa_medians <- DO_taxa |>
  # groups data by taxon so each species gets one median point
  group_by(taxon) |>
  # calculates median DO and median log abundance for each taxon
  summarise(
    # calculates the median DO value for each taxon
    med_do_point = median(med_do, na.rm = TRUE),
    # calculates the median log-transformed average abundance for each taxon
    med_log_ave_lit = median(log_ave_lit, na.rm = TRUE),
    # ungroup
    .groups = "drop")
```

```r
DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
                       # uses DO_taxa data frame
                       mapping = aes(x = med_do, # x-axis: med_do
                                     y = log_ave_lit, # y-axis: log_ave_lit
                                     color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_point(shape = 16, # shape is circle
             size = 2, # size is 2
             alpha = 0.7) + # opacity is 0.7
  # second layer: larger point shows the median location for each taxon
  geom_point(data = DO_taxa_medians,
             mapping = aes(x = med_do_point, # x value is median DO
                           # y value is median log-transformed abundance
                           y = med_log_ave_lit),
```

```
            shape = 18, # makes shape of points diamond
            size = 5) + # size of points is 5
  # separate panels for each taxon with free x- and y-axis scales
  facet_wrap(~ taxon, scales = "free") +
  # use conssitent colors for points
  scale_color_manual(values = taxon_colors) +
  labs(x = "Median DO (mg/L)", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
       # creates descriptive title
       title = "Invertebrate abundance as a function of DO") +
  # use consisten theme
  theme_ncos +
  # remove redundant legend
  theme(legend.position = "none")
# shows plot
DO_taxa_plot
```



Description of visual components

This figure shows the relationship between Median DO (mg/L) and log + 1 transformed average invertebrate abundance/L for our five focus taxa. Each panel represents a different taxon. Each small point represents one observation of that taxon, with median dissolved oxygen shown on the x-axis in mg/L and log + 1 transformed average abundance per liter shown on the y-axis. The different colors distinguish the taxa, and the larger diamond in each panel represents the overall average/summary point for that taxon across observations.

Main message / takeaway

Overall, there does not appear to be a strong or consistent relationship between median dissolved oxygen and invertebrate abundance across all taxa. Some taxa, especially Copepods, Ostracods, and Cladocerans, show higher abundances at low to moderate dissolved oxygen levels, but the pattern is variable and differs by taxon.

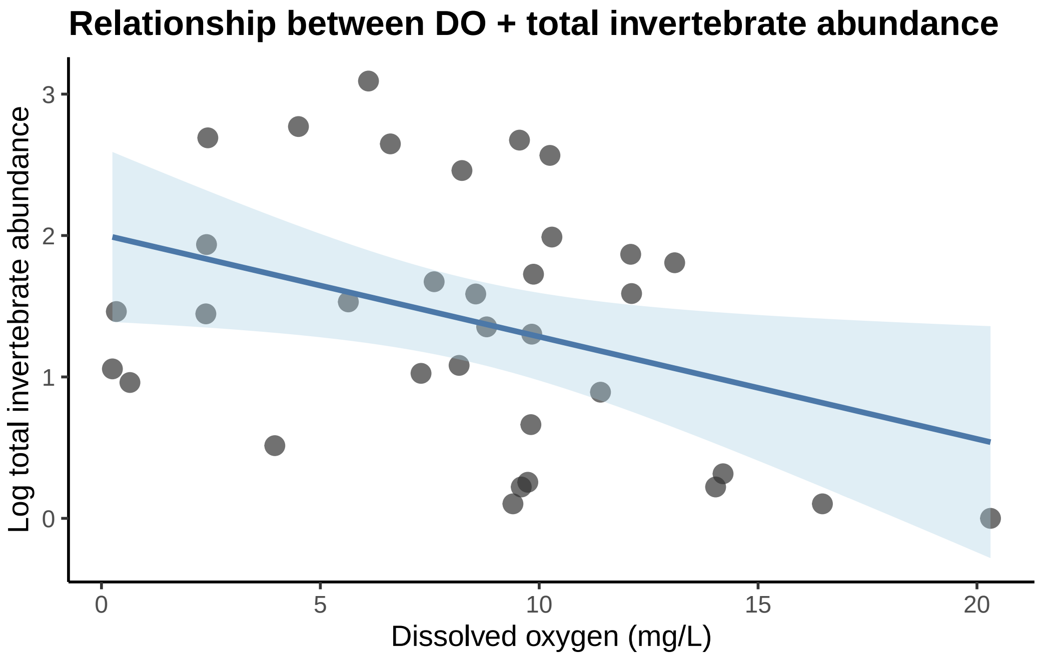
Connection to the main question

This figure helps evaluate whether dissolved oxygen is related to invertebrate abundance in the study system. The variation among taxa suggests that dissolved oxygen may influence some invertebrate groups differently, but DO alone does not clearly explain abundance patterns across the full invertebrate community.

Directly Relevant Figure 2: DO levels vs total invert. abundance

```
# fit linear model for relationship between log abundance and median DO
total_abundance_lm <- lm(log_total_abundance ~ med_do,
                         # use total_abundance as df
                         data = total_abundance)

# create new data frame total_abundance_predictions for predicted values
total_abundance_predictions <- data.frame(
  # create a sequence of DO values from the minimum to maximum observed DO
  med_do = seq(min(total_abundance$med_do, na.rm = TRUE), # max value is max DO
               max(total_abundance$med_do, na.rm = TRUE), # min value is min DO
               # creates 100 evenly spaced values for smooth predictions
               length.out = 100))

# generate predicted values and confidence intervals
total_abundance_predictions <- total_abundance_predictions |>
  bind_cols( # attach model predictions + confidence interval values to df
    # predict log total abundance using the fitted linear model
    predict(total_abundance_lm,
            # use the total_abundance_predictions df as data for prediction
            newdata = total_abundance_predictions,
            # create confidence intervals around the fitted values
            interval = "confidence") |>
      # convert prediction output into a data frame
      as.data.frame())
```

```r
# creat plot from total_abundance df
ggplot(total_abundance,
       aes(x = med_do, # x-axis is total_abundance
           y = log_total_abundance)) + # y-axis is log_total_abundance
  # layer 1: create points
  geom_point(color = raw_point_color, # color points
             size = 3, # size of points is 3
             alpha = 0.7) + # opacity of points is 0.7
  # layer 2: creates ribbon confidence interval around model predictions
  geom_ribbon(data = total_abundance_predictions,
              aes(x = med_do, # x is median DO
                  ymin = lwr, # min on y is lower confidence interval
                  ymax = upr), # max on y is lower confidence interval
              inherit.aes = FALSE, # does not inherit ggplot aesthetics
              fill = ci_fill, # colors ribbon
              alpha = 0.35) + # opacity of points is 0.35
  # layer 3: creates predicted line from linear model
  geom_line(data = total_abundance_predictions,
            aes(x = med_do, # x is median DO
                y = fit), # y is fitted model values
            inherit.aes = FALSE,  # does not inherit ggplot aesthetics
            color = model_color, # colors prediction line
            linewidth = 1) + # sets line width
  labs(
    x = "Dissolved oxygen (mg/L)", # labels x-axis
    y = "Log total invertebrate abundance", # labels y-axis
    # creates descriptive title
    title = "Relationship between DO + total invertebrate abundance") +
  # use consistent theme
  theme_ncos
```




```
# show the linear model results in a tidy table format
broom::tidy(total_abundance_lm)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic     p.value
  <chr>          <dbl>     <dbl>     <dbl>       <dbl>
1 (Intercept)   2.01      0.302       6.65 0.000000170
2 med_do       -0.0723    0.0316     -2.29 0.0289
```

Description of visual components

This figure shows the relationship between DO and total invertebrate abundance summed across our five focus taxa. Each pink point represents one sampling observation grouped by date, site, and median dissolved oxygen, where abundance was summed across the focus taxa and then log-transformed using log10(total_abundance + 1). The x-axis shows median dissolved oxygen in mg/L, and the y-axis shows log-transformed total invertebrate abundance. The purple line represents the predicted relationship from a linear model, and the light blue shaded area represents the confidence interval around the model prediction.

Main message / takeaway

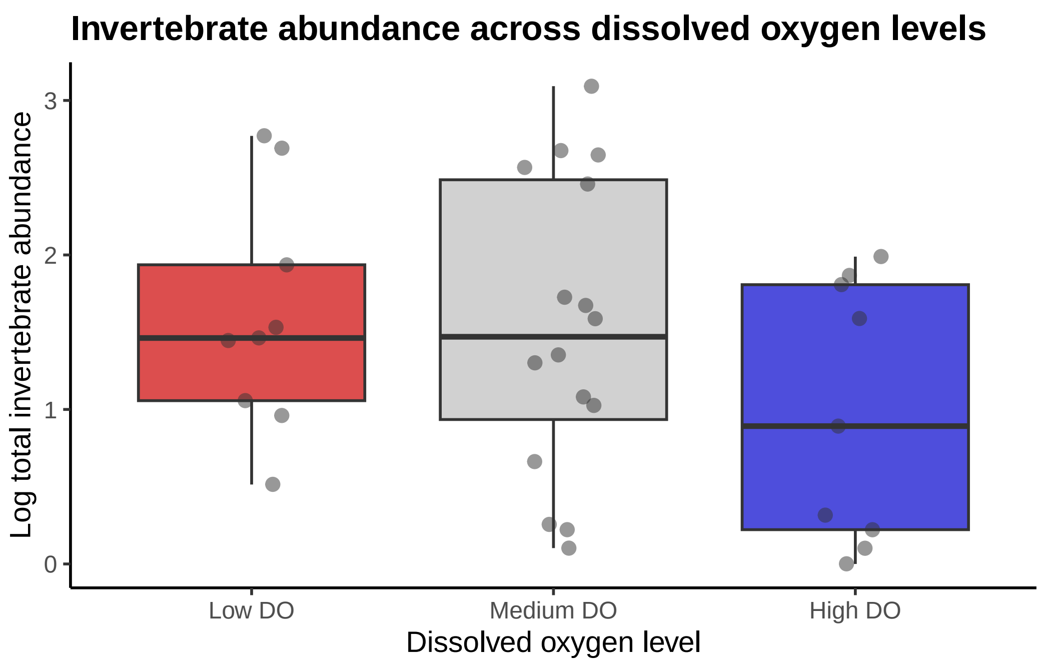
The figure shows a negative relationship between DO and total invertebrate abundance, where total invertebrate abundance tends to decrease as dissolved oxygen increases. However, there is still noticeable variation around the trend line, so dissolved oxygen explains some, but not all, of the variation in total abundance.

Connection to the main question

This figure directly addresses whether dissolved oxygen is related to invertebrate abundance by combining the focus taxa into one community-level abundance measure. It helps contextualize the main question by showing that lower dissolved oxygen conditions are associated with higher invertebrate abundance in this dataset.

Directly Relevant Figure 3: Abundance vs DO with binned levels

```
ggplot(total_abundance, # uses total_abundance df
       aes(x = do_category, # x-axis is do_category
           y = log_total_abundance, # y-axis is log_total_abundance
           fill = do_category)) + # fill color by do_category
  # layer 1: add boxplots to summarize abundance distribution w/in each DO level
  geom_boxplot(alpha = 0.7, # transparency is 0.7
               outlier.color = raw_point_color, # color outlier points
               outlier.size = 2) + # outlier size is 2
  # layer 2: adds jittered points of raw data
  geom_jitter(width = 0.15, # sets point width
              alpha = 0.5, # sets point opacity
              size = 2, # sets point size
              color = raw_point_color) + # sets point color
  labs(
    x = "Dissolved oxygen level", # labels x-axis
    y = "Log total invertebrate abundance", # labels y-axis
    # creates descriptive title
    title = "Invertebrate abundance across dissolved oxygen levels") +
  # manually assign fill colors to each DO category
  scale_fill_manual(values = c(
    # use red for low DO bin
    "Low DO" = 'red3',
    # use white for low mid bin
    "Medium DO" = 'grey',
    # use blue for low DO bin
    "High DO" = 'blue3')) +
  # uses consistent theme
  theme_ncos +
  # removes redundant legend
  theme(legend.position = "none")
```



Description of visual components

This figure compares log-transformed total invertebrate abundance across three DO categories: Low DO, Medium DO, and High DO. Each black jittered point represents one sampling observation grouped by date, site, and median dissolved oxygen, where abundance was summed across the five focus taxa and transformed using log10(total_abundance + 1). The colored boxes show the middle 50% of abundance values for each DO category, the thick horizontal line inside each box represents the median, and the whiskers show the spread of values outside the interquartile range.

Main message / takeaway

Total invertebrate abundance appears lower in the High DO category compared with the Low DO and Medium DO categories. However, there is substantial overlap and variability among the DO categories, especially in the Medium DO group, so the difference is not extremely clear-cut from the visualization alone.

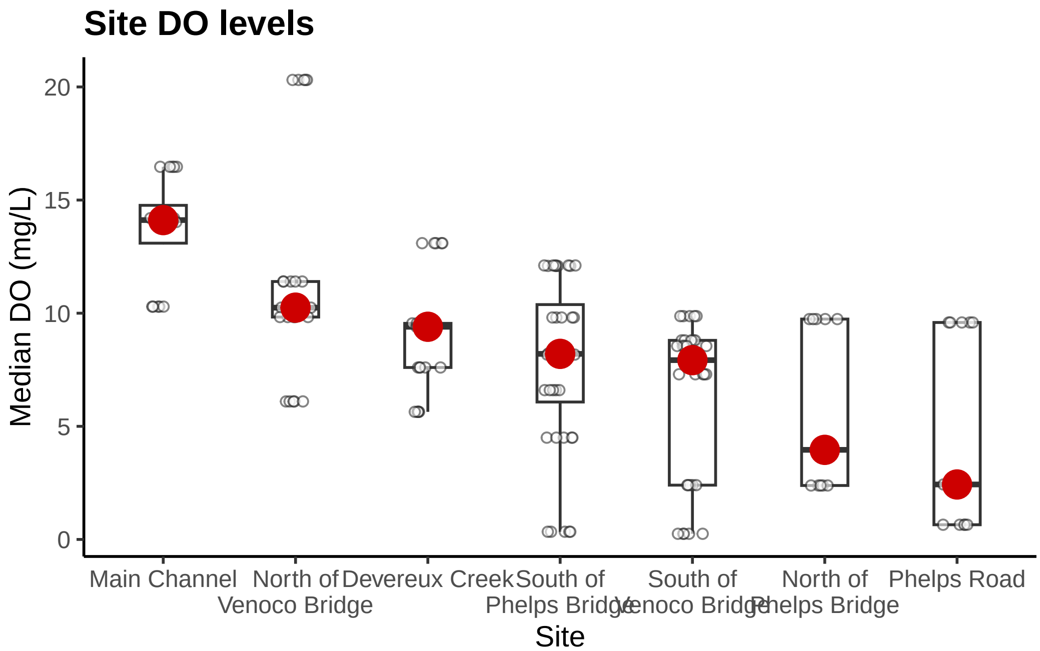
Connection to the main question

This figure helps address whether invertebrate abundance differs across dissolved oxygen levels by grouping continuous DO values into low, medium, and high categories. It shows that total invertebrate abundance may be higher under low-to-medium DO conditions and lower under high DO conditions in this dataset.

Other exploratory visualizations

Exploratory Figure 1: Variance of DO across sights

```
# create object aq_ins_clean_fig2 from aq_ins_clean
aq_ins_clean_fig2 <-  aq_ins_clean |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # filter to only include focus taxa
  filter(taxon %in% focus_taxa)

# base layer create plot from aq_ins_clean_fig2 df
DO <- ggplot(data = aq_ins_clean_fig2,
       mapping = aes(x = site_full, # x-axis: site
                     y = med_do)) + # y-axis: mean salinity
  # first layer: boxplot
  geom_boxplot(width = 0.35, outlier.shape = NA) +
  # second layer: jittered points for salinity measurements
  geom_jitter(height = 0, # sets height of jitter
              width = 0.12, # sets width of jitter
              color = raw_point_color, # colors jittered points
              shape = 21, # sets shape of jittered points
              fill = "white", # sets fill color
              alpha = 0.6) + # sets opacity of jittered points
  # wrapping x-axis text for readability
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent medians
  stat_summary(fun = median, # function is median value
               color = 'red3', # colors median points
               size = 1) + # sets size of median points
  labs(x = "Site", # labels x-axis
       y = "Median DO (mg/L)", # labels y-axis
       title = "Site DO levels") + # creates descriptive title
  # uses consistent theme
  theme_ncos

# shows plot
DO
```



Description of visual components

This figure shows how median DO varies across sampling sites. Each jittered open point represents one observation of median DO at a site, based on the focus invertebrate sampling data. The boxplots summarize the distribution of DO values at each site, where the box shows the IQR and the horizontal line inside the box shows the median. The large red point marks the median DO value for each site.

Main message / takeaway

There are clear differences in dissolved oxygen among sites. Main Channel has the highest DO values overall, while North of Phelps Bridge and Phelps Road tend to have the lowest DO values, with the other sites showing intermediate conditions.

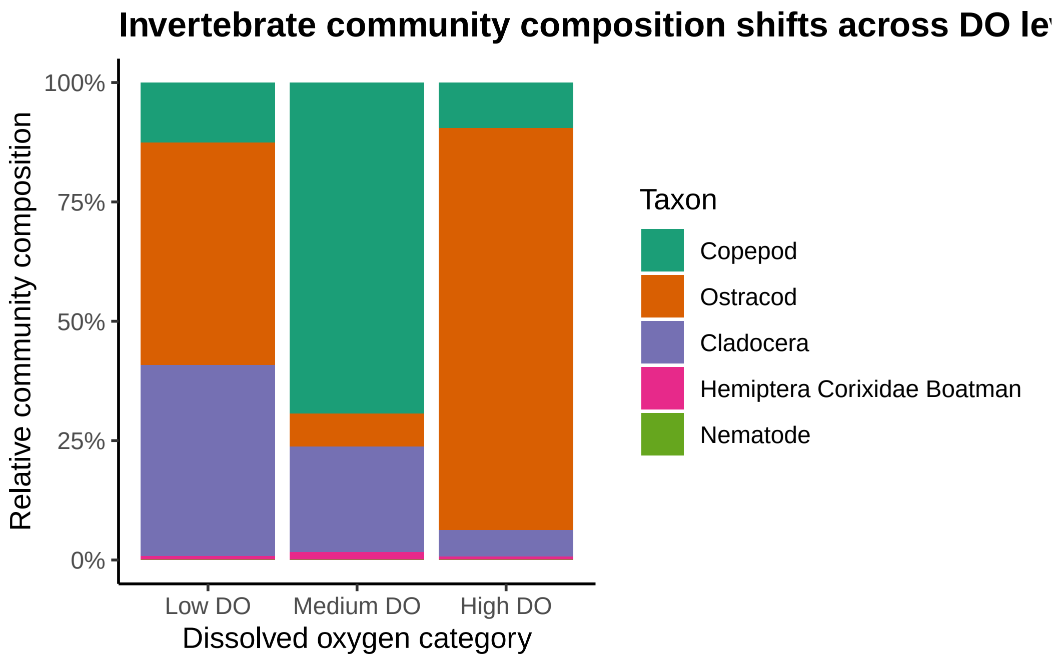
Connection to the main question

This figure helps contextualize the main question by showing that dissolved oxygen is not uniform across the study area, and varies substantially by site. These differences in DO between sites may help explain patterns in invertebrate abundance, since organisms at different sites are experiencing different oxygen conditions.

Exploratory Figure 2: Composition plot across DO bins

```r
# creates obejct composition_plot using DO_taxa df
composition_plot <- DO_taxa |>
  mutate(
    # classify each observation into low, medium, or high dissolved oxygen
    do_category = case_when(
      # assign Low DO to values at or below the 25th percentile of median DO
      med_do <= quantile(med_do, 0.25, na.rm = TRUE) ~ "Low DO",
      # assign Medium DO to values between the 25th percentile - 75th percentile
      med_do <= quantile(med_do, 0.75, na.rm = TRUE) ~ "Medium DO",
      # assign High DO to all remaining values above the 75th percentile
      TRUE ~ "High DO"),
    # convert do_category into a factor to control plotting order
    do_category = factor(do_category,
                         # set order of DO categories on the x-axis
                         levels = c("Low DO", "Medium DO", "High DO"))) |>
  # groups by do_category and taxon
  group_by(do_category, taxon) |>
  # calculate total abundance for each taxon within each DO category
  summarize(
    # sum average abundance per liter + ignore missing values
    total_abundance = sum(ave_lit, na.rm = TRUE),
    # ungroup
    .groups = "drop") |>

# create plot from composition_plot df
ggplot(aes(x = do_category, # x-axis is do_category
           y = total_abundance, # y-axis is total_abundance
           fill = taxon)) + # fill color by taxon
  # create stacked bars scaled to proportions within each DO category
  geom_col(position = "fill") +
  # use consistent colors
  scale_fill_manual(values = taxon_colors) +
  # format the y-axis as percentages
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    x = "Dissolved oxygen category", # relabels x-axis
    y = "Relative community composition", # relabels y-axis
    fill = "Taxon", # labels legend
    title = "Invertebrate community composition shifts across DO levels") +
  # use consistent theme
  theme_ncos
```

```
composition_plot
```



Description of visual components

This figure shows how relative invertebrate community composition changes across three dissolved oxygen categories: Low DO, Medium DO, and High DO. Each stacked bar represents the total abundance of the focus taxa within one DO category, scaled to 100% so that the bar shows relative composition rather than raw abundance. Each colored segment within a bar represents the proportion contributed by one focus taxon and the y-axis shows the percent of the community made up by each group.

Main message / takeaway

The figure suggests that invertebrate community composition shifts across dissolved oxygen levels. Low DO sites have a more mixed community with large contributions from Cladocera and Ostracods, Medium DO sites are dominated by Copepods, and High DO sites are dominated primarily by Ostracods, while Hemiptera Corixidae Boatman and Nematodes remain minor components across all DO levels.

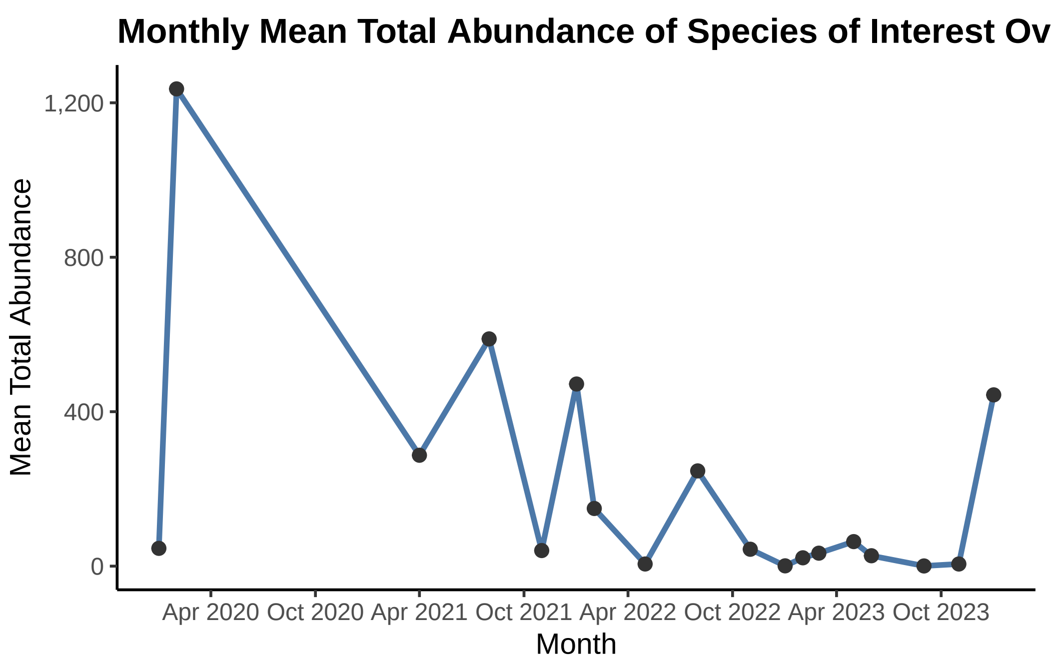
Connection to the main question

This figure helps address the main question by showing that dissolved oxygen may influence not only total invertebrate abundance, but also which taxa make up the community. It provides important context by suggesting that changes in oxygen conditions are associated with shifts in community structure, meaning different invertebrate groups may respond to different DO levels.

Exploratory Figure 3: Time-series plot of total abundance (all species)

```
# create total abundance over time
total_abundance_time <- DO_taxa |>
  # keeps only the taxa of interest
  filter(taxon %in% focus_taxa) |>
  # makes sure date is treated as a date object
  mutate(date = as.Date(date)) |>
  # creates a month column so close sampling dates are summarized together
  mutate(month = lubridate::floor_date(date, unit = "month")) |>
  # groups by date and month
  group_by(date, month) |>
  # calculates total abundance across sites and taxa for each sampling date
  summarise(
    # sums average abundance per liter across all focus taxa and sites
    daily_total_abundance = sum(ave_lit, na.rm = TRUE),
    # ungroup
    .groups = "drop") |>
  # groups daily abundance totals by month
  group_by(month) |>
  # calculates one summarized abundance value for each month
  summarise(
    # calculates mean total abundance for each month
    mean_total_abundance = mean(daily_total_abundance, na.rm = TRUE),
    # counts how many sampling dates are included in each monthly value
    n_sampling_dates = n(),
    # ungroup
    .groups = "drop")

# plot total abundance over time
ggplot(total_abundance_time,
       aes(x = month, # x-axis is month
           y = mean_total_abundance)) + # y-axis is mean_total_abundance
  # layer 1: adds a line connecting total abundance values over time
  geom_line(linewidth = 1, # sets line width
            color = model_color) + # sets line color
  # layer 2: adds points for each sampling date
  geom_point(size = 2, # sets point size
```

```
            color = raw_point_color) + # sets point color
  # formats y-axis labels with commas
  scale_y_continuous(labels = scales::comma) +
  # formats the x-axis to show dates clearly
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
  # adds clear labels
  labs(# creates descriptive title
    title = "Monthly Mean Total Abundance of Species of Interest Over Time",
    x = "Month", # labels x-axis
    y = "Mean Total Abundance") + # labels y-axis
  # uses consistent theme ***NEED TO UPDATE TITLE FOR READABILITY***
  theme_ncos
```



Clarification: To reduce visual clutter from sampling dates that occurred close together, total abundance was first calculated for each sampling date and then averaged within each month. This monthly summary makes broader temporal patterns easier to interpret than plotting every individual sampling date.

Description of visual components

This figure shows total abundance of the focus invertebrate taxa over time. Each black point represents one sampling date, where abundance was summed across all sites and across all taxa of interest. The green line connects sampling dates in chronological order to show changes in total abundance over time. The x-axis shows sampling date, and the y-axis shows total abundance.

Main message / takeaway

Total invertebrate abundance varies substantially over time, with one very large peak in 2020 and several smaller peaks in later years. There is no consistent increasing or decreasing trend across the full time period. Abundance appears highly variable among sampling dates.

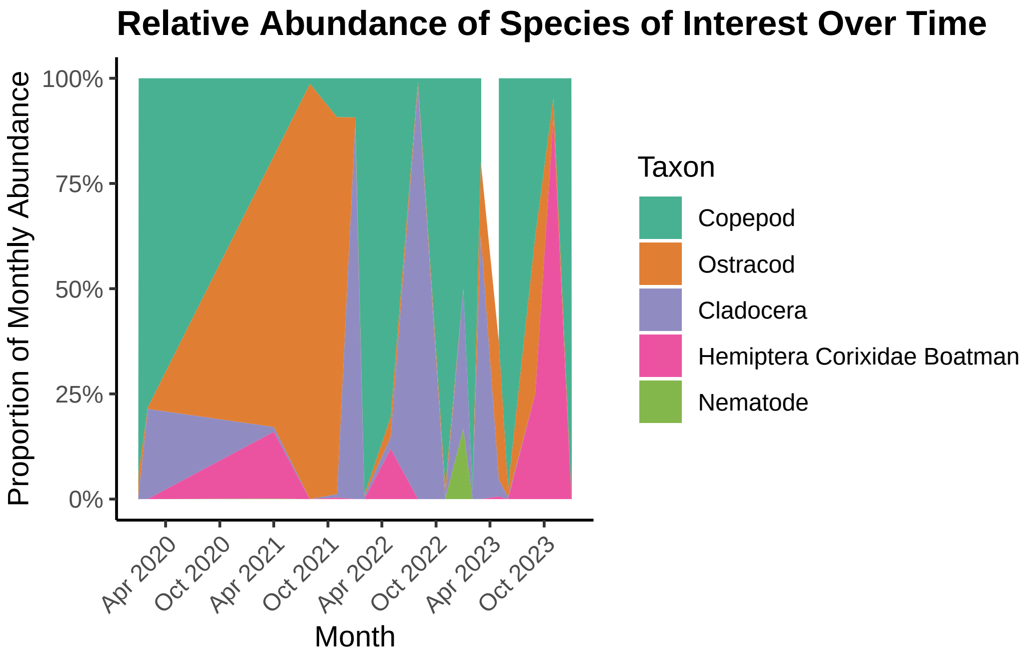
Connection to the main question

This figure provides context for the dissolved oxygen and invertebrate abundance relationship by showing that abundance changes strongly across sampling dates. This is important because variation over time may influence observed abundance patterns in addition to dissolved oxygen levels.

Exploratory Figure 4: Time-series area plot

```
# create object taxon_abundance_time from DO_taxa
taxon_abundance_time <- DO_taxa |>
  # keeps only the focus taxa
  filter(taxon %in% focus_taxa) |>
  # makes sure date is treated as a Date object
  mutate(date = as.Date(date)) |>
  # creates a month column
  mutate(month = lubridate::floor_date(date, unit = "month")) |>
  # groups by date, month, and taxon
  group_by(date, month, taxon) |>
  # sums abundance across sites for each taxon on each sampling date
  summarise(
    # calculates total abundance per taxon for each sampling date
    daily_taxon_abundance = sum(ave_lit, na.rm = TRUE),
    # ungroup
    .groups = "drop") |>
  # groups by month and taxon
  group_by(month, taxon) |>
  # calculates monthly mean abundance for each taxon
  summarise(
    # averages taxon abundance across sampling dates within each month
    monthly_taxon_abundance = mean(daily_taxon_abundance, na.rm = TRUE),
    # ungroup
    .groups = "drop") |>
  # groups by month
```

```r
  group_by(month) |>
  # calculates each taxon's proportion of total monthly abundance
  mutate(# divides each taxon's monthly abundance by the total abundance per month
    proportion = monthly_taxon_abundance / sum(monthly_taxon_abundance)) |>
  # ungroup
  ungroup()

# create time series area plot using taxon_abundance_time df
ggplot(taxon_abundance_time,
       aes(x = month, # x-axis is month
           y = proportion, # y-axis is proportion
           fill = taxon)) + # fill color by taxon
  # creates stacked area plot where total height is 100% of abundance per month
  geom_area(alpha = 0.8) + # sets opacity
  # uses consistent colors
  scale_fill_manual(values = taxon_colors) +
  # formats y-axis as percentages and keeps limits from 0% to 100%
  scale_y_continuous(labels = scales::percent_format(), limits = c(0, 1)) +
  # formats x-axis dates so monthly time patterns are easier to read
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
  # adds clear labels
  labs(
    # creates descriptive title
    title = "Relative Abundance of Species of Interest Over Time",
    x = "Month", # labels x-axis
    y = "Proportion of Monthly Abundance", # labels y-axis
    fill = "Taxon") +# labels fill legend
  # uses consistent theme
  theme_ncos +
  # rotates x-axis text to make month labels easier to read
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



To make the time series easier to interpret, abundance was summarized by month rather than plotted for every individual sampling date. The stacked area plot shows the proportional contribution of each taxon to total monthly abundance, so the y-axis ranges from 0% to 100% and highlights changes in community composition over time.

Description of visual components

This figure shows changes in abundance over time for our five focus invertebrate taxa: Ostracod, Copepod, Cladocera, Hemiptera Corixidae Boatman, and Nematode. Each different colored area represents the summed average abundance per liter for one taxon on each sampling date, summed across sites but kept separate by taxon. The stacked height of the colored areas represents the total abundance of all focus taxa combined at that sampling date. The x-axis shows sampling date, and the y-axis shows average abundance per liter.

Main message / takeaway

Abundance varies strongly over time, and different taxa contribute to peaks at different points. Copepods appear to drive the largest peak in 2020 and another increase near 2024, while Ostracods and Cladocera contribute noticeably to some intermediate peaks.

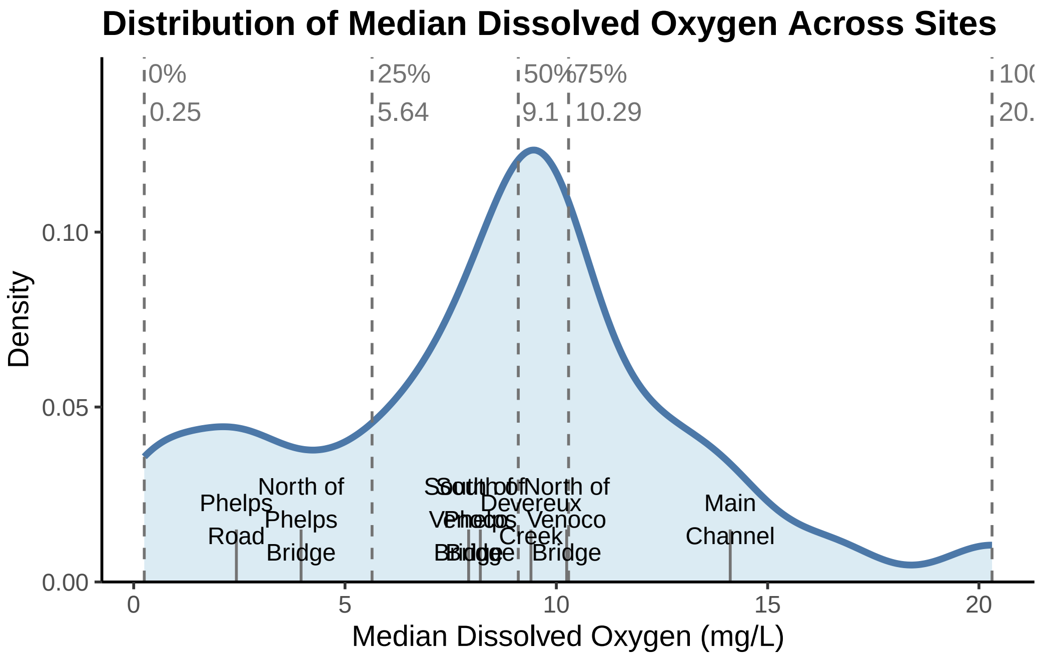
Connection to the main question

This figure adds time and taxon-specific context to the relationship between dissolved oxygen and invertebrate abundance. It shows that abundance patterns are not constant over time and that changes in total abundance may be driven by shifts in particular taxa, which is important when interpreting how the community responds to DO conditions.

Exploratory Figure 5: Distribution of Median Dissolved Oxygen Aross Sites

```
# create new object median_fig2 from aq_ins_clean_fig2
median_fig2 <- aq_ins_clean_fig2 |>
  # group by site_full
  group_by(site_full) |>
  # calculate the median DO value for each site
  summarize(median_do = median(med_do, na.rm = TRUE))

# create new column in aq_ins_clean_fig2 that assigns quartiles for each site
aq_ins_clean_fig2$quartile <- cut(
  # use median DO values to assign quartile categories
  aq_ins_clean_fig2$med_do,
  # define quartile breaks using the 0%, 25%, 50%, 75%, and 100% quantiles
  breaks = quantile(aq_ins_clean_fig2$med_do, probs = seq(0, 1, 0.25),
                    na.rm = TRUE), # discludes NA values
  # label each quartile group
  labels = c("Q1", "Q2", "Q3", "Q4"),
  # include the lowest DO value in the first quartile interval
  include.lowest = TRUE,
  # make intervals left-closed and right-open
  right = FALSE)

# boundaries of quartiles
q <- quantile(aq_ins_clean_fig2$med_do, probs = seq(0, 1, 0.25), na.rm = T)
# create a density object for median dissolved oxygen values
dens <- density(aq_ins_clean_fig2$med_do, na.rm = TRUE)

# create new object label_df from aq_ins_clean_fig2
label_df <- aq_ins_clean_fig2 |>
  # group the data by site so each site has one label location
  group_by(site_full) |>
  # calculate the median DO value for each site
  summarize(median_do = median(med_do, na.rm = TRUE))

# approximate density height for each site
label_df$y <- approx(
  dens$x, # use the x-values from the density curve
  dens$y, # estimate density height at each site's median DO value
  xout = label_df$median_do)$y

# distribution plot of DO levels across sites
```

```r
ggplot(aq_ins_clean_fig2, aes(x = med_do)) +
  # layer 1: density curve
  geom_density(fill = ci_fill, # determines color of fill of curve
               alpha = 0.4, # determines opacity
               color = model_color, # determines color of line of curve
               linewidth = 1.2) + # determines line width of curve
  # layer 2: vertical quartile lines
  geom_vline(xintercept = q,
             linetype = "dashed", # make quartile lines dashed
             color = reference_color) + # set quartile line color
  # segments pointing to sites
   geom_segment(data = label_df, # use label_df data for site median markers
                aes(x = median_do, # x-position is site median DO
                    xend = median_do, # keep ending x-position at site median DO
                    y = 0, # start segment at bottom of plot
                    yend = 0.015), # end segment slightly above bottom of plot
                color = reference_color) + # set segment color
  # create site labels
  geom_text(data = label_df, # use label_df data
            aes(x = median_do, # place site labels at each site's median DO value
                y = 0.018, # sets vertical position for site labels
                # wrap long site names for readability
                label = str_wrap(site_full, width = 10)),
            size = 3) + # sets site label text size
  # creates quartile dashed line labels
  geom_text(data = data.frame(x = q, label = names(q)),
            # label each quartile line with its name and rounded DO value
            aes(x = x, y = 0.14, label = paste0(label, "\n", round(x, 2))),
            color = reference_color, # sets quartile label color
            size = 3.5, # sets quartile size color
            hjust = -0.1) + # shift text slightly to the right of the line
  # remove space between bottom of plot and y-axis, set y-axis limits
  scale_y_continuous(expand = c(0,0), limits = c(0, 0.15)) +
  # axes labels and title
  labs(x = "Median Dissolved Oxygen (mg/L)", # labels x-axis
       y = "Density", # labels y-axis
       # creates descriptive title
       title = "Distribution of Median Dissolved Oxygen Across Sites") +
  # uses consistent theme
  theme_ncos
```




```r
# removes legend
theme(legend.position = "none")
```

```
List of 1
 $ legend.position: chr "none"
 - attr(*, "class")= chr [1:2] "theme" "gg"
 - attr(*, "complete")= logi FALSE
 - attr(*, "validate")= logi TRUE
```

These bins were created by according to the median dissolved oxygen levels at each site. Low DO is the 25th percentile, moderate DO is the 75th percentile, and high DO is the 100th percentile. The labels show the median of the median DO measurments at each site.

- Low DO: Phelps Road and North of Phelps Bridge
- Moderate DO: South of Venoco Bridge, South of Phelps Bridge, Devereux Creek, North of Venoco Bridge
- High DO: Main Channel

Description of visual components

This figure shows the distribution of median dissolved oxygen values across the invertebrate sampling observations. The blue density curve shows where DO values are most concentrated, with higher areas of the curve indicating more observations at those DO levels. The dashed vertical lines mark the minimum, 25th percentile, median, 75th percentile, and maximum DO values, which are used to understand the spread of DO conditions. The site labels and short vertical gray lines near the bottom show the median DO value for each site. This allows site-level DO conditions to be easily compared.

Main message / takeaway

Median DO values are mostly concentrated around roughly 8–11 mg/L, but there is a wide overall range from very low DO values to values above 20 mg/L. Sites differ in their typical DO levels, with Phelps Road and North of Phelps Bridge having relatively low median DO, while Main Channel has a higher median DO.

Connection to the main question

This figure helps contextualize the main question by showing the range of dissolved oxygen conditions that invertebrates experienced across sites. It also supports the use of low, medium, and high DO categories by showing where the quartile cutoffs occur within the DO distribution.

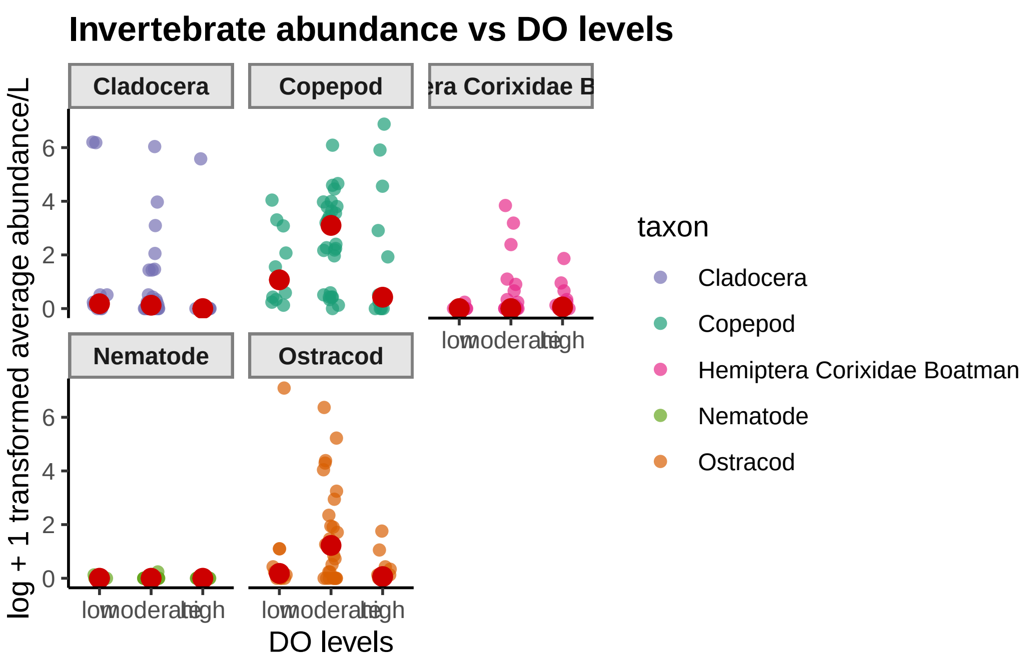
Exploratory Figure 6: Invertebrate Abundance in DO Bins

```
# base layer: ggplot
ins_bins_plot <- ggplot(data = aq_ins_clean_bins, # uses aq_ins_clean_bins df
                        mapping = aes(x = bins, # x-axis: DO bins
                                      y = log_ave_lit, # y-axis: log_ave_lit
                                      color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_jitter(width = 0.15, # sets jitter width
              height = 0, # sets jitter height
              shape = 16, # sets jitter point shape
              size = 2, # sets jitter point size
              alpha = 0.7) + # sets jitter point opacity
  # calculates median for each bin for each taxon
  stat_summary(fun = median, # uses median function
               geom = "point", # uses point geometry
               color = "red3", # colors median points red
               size = 3) + # sets median point size
  # creates separate panels for each taxon
  facet_wrap(~ taxon) +
  # uses consistent colors for points
  scale_color_manual(values = taxon_colors) +
  labs(x = "DO levels", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
```

```
      # creates descriptive title
      title = "Invertebrate abundance vs DO levels") +
  # uses consistent theme
  theme_ncos
  # remove redundant legend
  theme(legend.position = "none")
```

```
List of 1
 $ legend.position: chr "none"
 - attr(*, "class")= chr [1:2] "theme" "gg"
 - attr(*, "complete")= logi FALSE
 - attr(*, "validate")= logi TRUE
```

```
# shows plot
ins_bins_plot
```



Description of visual components

This figure shows taxon-specific invertebrate abundance across the three dissolved oxygen categories: low, moderate, and high. Each panel represents one focus taxon. Each small colored point represents one sampling observation for that taxon, with the x-axis showing the DO category and the y-axis showing log + 1 transformed average abundance per liter. The large red points represent the median abundance for each taxon within each DO category.

Main message / takeaway

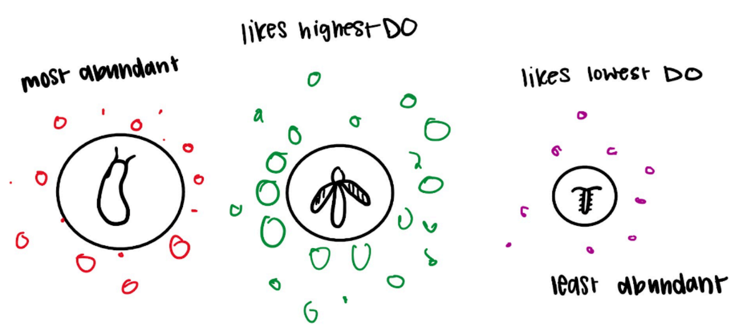
The figure suggests that the relationship between abundance and DO level differs among taxa rather than following one single community-wide pattern. Copepods and Ostracods tend to show higher and more variable abundances in the moderate DO category, while Nematodes remain near zero across all DO levels and Hemiptera Corixidae Boatman abundace stays generally low with only a few higher values at moderate DO.

Connection to the main question

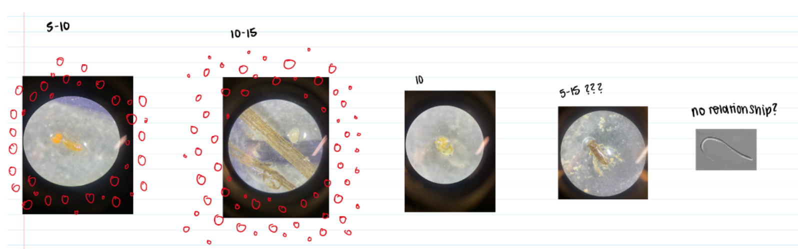
This figure helps address the main question by showing whether individual invertebrate taxa respond differently across dissolved oxygen conditions. It adds important context by demonstrating that DO effects may be taxon-specific, meaning patterns in total abundance could be driven by stronger responses in some groups than in others.

Plan for elective

Draft 1:

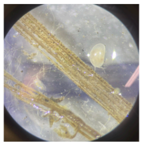
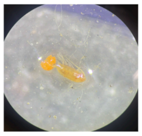
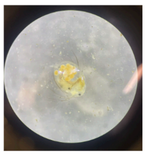


Draft 2:



Focus Taxa Images:

Most common inverts

1. Ostracod
   a. 
2. Copepod
   a. 
3. Cladocera
   a. 
4. Corixidae (Water Boatmen)
   a. 
5. Nematode

Elective idea general concept: Our elective aims to show several factors we examined for the relationship between NCOS aquatic inverts and dissolved oxygen. Using the size of the microscopic images of each of the 5 main invert species, we are able to display abundance. The larger the picture, the more abundant that species is. We then plan to paint watercolor bubbles around each invert to show each species preferred DO levels.

Progress we have made: Each of the microscopic images were collected through samples at the NCOS lab (with the exception of nematodes) for this elective. Another large area of progress that has been made is actually figuring out what the relationship between each invert and DO is. We found that there is little to no correlation between these two factors and that each invert prefers similar low levels of DO.

Possible changes that should be made based on our findings: Rather than bubbles representing each species proffered amount of DO, the bubbles could represent the strength of the relationship between the species abundance and the DO levels. For species with a negative correlation, we could draw the bubbles a different color.